

Time-Aware CP-ABE with Threshold Policies for Digital Twin Security in Industrial IoT

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Abstract—Existing ciphertext-policy attribute-based encryption schemes for industrial IoT digital twins lack native temporal access control, flexible threshold policies, and efficient attribute revocation in a unified framework, leaving systems vulnerable to time-based and privilege-escalation attacks. We propose T-CP-ABE, a time-aware CP-ABE framework that natively integrates k -of- n THRESHOLD gates and version-based attribute revocation with forward security: temporal access control is enforced through LSSS policy attributes with hash-chain forward security, while BLS signatures and a conditional diffusion model provide complementary authentication and anomaly detection layers. We formalize the IND-sCPA-T security model—a security proof that incorporates both temporal token queries and attribute-version revocation queries within a unified adversary model—and prove security under the DBDH assumption. Implemented in Charm-Crypto, the framework achieves sub-300 ms decryption for typical IoT policies (≤ 10 attributes) with key-generation performance stable across 10–10,000 user attributes, and linear policy scaling (23.6 s at 1000 policy attributes) acceptable for offline batch audit. Validation on UNSW-NB15 and four real-world scenarios confirms practical applicability for industrial digital twin deployments.

Index Terms—Attribute-based encryption, time-aware access control, digital twin, industrial IoT, threshold policy

I. INTRODUCTION

RECENTLY, the convergence of digital twin technology and the Industrial Internet of Things (IIoT) has markedly transformed manufacturing, healthcare, and smart city infrastructure through virtual replicas of physical assets that enable real-time monitoring, predictive maintenance, and lifecycle management [1]–[3]. As these deployments scale across thousands of heterogeneous devices and organizational boundaries, securing access to twin data becomes critical—unauthorized reading exposes industrial secrets, while unauthorized modification can cascade into physical-world safety failures. Ciphertext-policy attribute-based encryption (CP-ABE) embeds access policies directly in ciphertexts and allows decryption only by users whose attributes satisfy the policy, offering a natural fit for fine-grained access control in such environments.

However, practical deployment of CP-ABE in IIoT digital twin scenarios is hindered by the absence of three features that must operate in concert: temporal access control, flexible threshold policies, and efficient attribute revocation [4]–[7]. Existing schemes address these in isolation—Zhu et al. [8] established temporal access control via cryptographic integer

comparisons, Liu et al. [9] combined time-based direct revocation with CP-ABE, and Hur et al. [10] tackled attribute revocation—yet no construction unifies all three. Combining these mechanisms introduces interaction-induced attack vectors absent when each is analyzed in isolation: a revoked-but-unexpired time token could forge a decryption path unless version tags are checked alongside the hash chain; injecting time tokens as additional Shamir shares would alter the polynomial degree and lower the threshold, so time metadata must be bound orthogonally to the share structure; an adversary holding $k-1$ shares, a valid time token, and a revoked key could bypass version checks if the integration is not carefully designed. These challenges demand a combined construction with a formal security model capturing the joint behavior of all mechanisms.

To reconcile this tension, we propose T-CP-ABE, a time-aware CP-ABE framework for digital twin security in IIoT. The contributions are as follows. We present a construction that natively supports THRESHOLD(k, n) gates and version-based attribute revocation with forward security, where temporal access control is enforced through time predicate rows in the LSSS matrix with hash-chain-based forward security, and a publicly computable metadata tag records the encryption period for auditability. We formalize the IND-sCPA-T security model—a selective chosen-plaintext game extending standard IND-sCPA with time token and revocation queries—and prove security under the DBDH assumption with advantage bound $\text{Adv} \leq 2\text{Adv}^{\text{DBDH}} + (q_H + q_K + q_T)/p$. This security proof incorporates both temporal token queries and attribute-version revocation queries within a unified adversary model, capturing interaction-induced attack vectors absent when each mechanism is analyzed in isolation. We further integrate BLS digital signatures [11] for mutual authentication and a conditional diffusion model [12]–[14] for anomaly detection as complementary security layers. Evaluation using Charm-Crypto demonstrates sub-300 ms decryption for typical IoT policies, stable performance scaling to 10,000 attributes, and validation on the UNSW-NB15 dataset [15] alongside four real-world deployment scenarios.

The remainder of this article is organized as follows. Section II reviews related work on CP-ABE, temporal access control, and digital twin security. Section III presents the mathematical preliminaries. Section IV defines the system architecture, threat model, and IND-sCPA-T security game. Section V describes the T-CP-ABE construction and its security proofs.

Section VI reports experimental results. Section VII concludes with limitations and future directions.

II. RELATED WORK

Sahai and Waters [16] pioneered ABE by linking attributes to keys and access structures to ciphertexts [17]. Bethencourt et al. [18] realized the ciphertext-policy variant (CP-ABE), and Waters [19] achieved full non-selective security via linear secret sharing schemes. Subsequent work addressed decryption outsourcing, multi-authority constructions [20]–[23], attribute revocation [24], [25], pairing-free constructions for resource-constrained IoT devices [26], [27], and lattice-based post-quantum ABE [4], [28]. However, these schemes do not support temporal access control, and cloud-based offloading remains impractical for the massive data volumes generated by IIoT devices.

Zhu et al. [8] established temporal access control by combining cryptographic integer comparisons with CP-ABE, enabling time-bound encryption without a trusted clock. Liu et al. [9] extended this with time-based direct revocation, though without flexible threshold policies. Hu et al. [29] proposed a time-attribute-based access control scheme for SDN-IoT environments. Cai et al. [30] and Li et al. [31] further explored time-bound ABE with outsourcing support. All treat time as an external constraint rather than an integral part of the access structure, and neither provides forward security through hash chains. Combining temporal predicates with revocation introduces interaction-induced attack vectors—a revoked-but-unexpired token could forge a decryption path unless version tags are checked alongside the hash chain—yet no existing scheme addresses these interactions with a formal security model.

Digital twin security frameworks [3], [32]–[34] address encryption and integrity but lack fine-grained, time-aware access control. Dai et al. [35] combined blockchain with CP-ABE for digital twin sharing without temporal constraints or forward security. Hybrid AI-driven access control frameworks have been proposed for IIoT environments [17], yet runtime threat detection via machine learning [36] operates independently of cryptographic enforcement, leaving a gap between detection and access control [5], [37]. Recent AI-driven security frameworks—Khan et al. [38] proposed a hybrid ABAC with ML-driven policy adjustment for IIoT, and Rahman et al. [39] integrated ABAC with federated learning for smart homes—but neither closes the loop from anomaly detection to cryptographic policy adaptation. We also acknowledge the limitations of our pairing-based construction: pairing-free schemes [40] achieve lower decryption cost but require trusted proxy assumptions incompatible with our threat model; lattice-based post-quantum ABE [41], [42] remains impractical for resource-constrained IIoT due to key and ciphertext size overhead exceeding 10×; and multi-authority extensions [23], [43], [44] eliminate single-TA key escrow but introduce coordination latency unsuitable for real-time twin synchronization. These trade-offs define the applicability boundary of T-CP-ABE: centralized IIoT deployments where temporal access control and threshold policies are primary

TABLE I
COMPARISON OF CP-ABE SCHEMES.

Scheme	Time	Threshold	Revoc.	Sec.
BSW07 [18]	×	×	×	Sel.
Guo24 [26]	×	×	×	Sel.
Zhang24 [27]	×	×	✓	Sel.
Gasparini26 [28]	×	×	✓	Sel.
Chen25 [40]	×	×	×	Sel.
Poomekum25 [41]	×	×	✓	Sel.
T-CP-ABE (Ours)	✓	✓	✓	Sel.

requirements. To address these gaps, we propose T-CP-ABE, which unifies temporal predicates, THRESHOLD gates, and version-based revocation in a single CP-ABE construction.

Compared with schemes addressing a subset of these requirements, T-CP-ABE provides the following distinctions. W.r.t. ABE with revocation [24], [25], we additionally support temporal predicates and THRESHOLD gates natively in the LSSS matrix. W.r.t. time-bound ABE [8], [9], [29], we integrate version-based attribute revocation with forward security. W.r.t. multi-authority ABE [21], [23], [43], [44], our single-TA design trades decentralization for simplicity, suitable for digital twin deployments with a centralized trust anchor. W.r.t. pairing-free ABE [40], our pairing-based construction targets IIoT devices with moderate computational capacity where trusted proxy assumptions are unacceptable. W.r.t. lattice-based post-quantum ABE [28], [41], [42], our pairing-based construction targets IIoT devices with moderate computational capacity and does not address quantum adversaries—an acknowledged limitation with concrete paths forward (Section VII).

T-CP-ABE integrates BLS-based timestamp authentication and AI-driven anomaly detection within a single framework. Poomekum25 and Gasparini26 achieve post-quantum security via lattice-based construction.

III. PRELIMINARIES

A. Bilinear Pairings

Let $\mathcal{P} = (\mathbb{G}, \mathbb{G}_T, p, e, g)$ be a bilinear group of prime order p [45] over the SS1024 curve (≈ 128 -bit security).

B. Linear Secret Sharing Schemes

An LSSS over $\mathbb{Z}_p$ encodes a secret s via an access structure $\mathbb{A} = (\mathbb{M}, \rho)$, where $\mathbb{M} \in \mathbb{Z}_p^{\ell \times n}$ and ρ maps row i to an attribute. A dealer selects $\vec{v} = (s, v_2, \dots, v_n) \in_R \mathbb{Z}_p^n$ and assigns shares $\lambda_i = \mathbb{M}_i \cdot \vec{v}$. For THRESHOLD(k, n) gates [46], any k of n shares recover s via Lagrange interpolation:

$$s = \sum_{j \in S} \lambda_j \cdot \Delta_{j,S}(0), \quad \Delta_{j,S}(0) = \prod_{i \in S, i \neq j} \frac{0 - i}{j - i} \pmod{p} \quad (1)$$

C. Complexity Assumptions

Our complexity assumption is as follows.

Given a bilinear group tuple $(\mathbb{G}, \mathbb{G}_T, p, e)$ generated by $\mathcal{J}(\lambda)$ with generator $g \in \mathbb{G}$, and random values $a, b, c \in_R \mathbb{Z}_p$

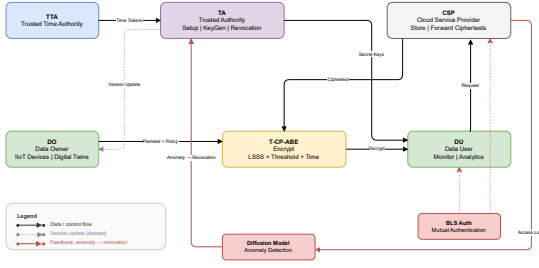


Fig. 1. System architecture of the time-aware digital twin security framework.

chosen by a challenger $\mathcal{C}$, the DBDH assumption posits that no probabilistic polynomial-time (PPT) algorithm $\mathcal{B}$ can distinguish between $T = e(g, g)^{abc}$ and $T \in_R \mathbb{G}_T$ with non-negligible advantage, where $\vec{g} = (g, g^a, g^b, g^c)$. Formally,

$$\text{Adv}_{\mathcal{B}}^{\text{DBDH}}(\lambda) = |\Pr[\mathcal{B}(\vec{g}, e(g, g)^{abc}) = 1] - \Pr[\mathcal{B}(\vec{g}, T_{\text{rand}}) = 1]| \quad (2)$$

IV. SYSTEM DEFINITIONS AND SECURITY MODEL

A. System Architecture

Fig. 1 illustrates the four entities of the T-CP-ABE system.

1) **TA**: Responsible for system setup and key generation. Fully trusted; multi-authority extensions [20]–[22] can mitigate single-point-of-failure risk but are out of scope.

2) **CSP**: Stores encrypted digital twin data and forwards ciphertexts to DUs. Semi-trusted—follows the protocol but may attempt to learn from ciphertexts.

3) **DOs**: Deploy IIoT devices and digital twins, set access policies, and issue attribute revocation updates.

4) **DUs**: Stakeholders who access digital twin data for monitoring or analytics. They authenticate via BLS signatures and decrypt ciphertexts whose policy their attributes satisfy and whose time token falls within the authorized window.

Data flows from DO-managed devices through T-CP-ABE encryption with temporal predicates and THRESHOLD gates to the CSP, which forwards ciphertexts only to authorized DUs. Attribute version updates ensure revoked users cannot access new data. The diffusion model [13], [14], [47] computes anomaly scores on access requests, triggering dynamic policy tightening when scores exceed the 95th-percentile threshold.

B. Threat Model

The adversary is a PPT algorithm that can observe public parameters and ciphertexts, corrupt any subset of users as long as the corrupted set does not satisfy the target access structure, and issue chosen-plaintext queries. The adversary cannot compromise the TA or forge time tokens. Time tokens T_i are publicly released at each period i and are not secret. The binding factor $b = \mathcal{H}_Z(i \| T_i) \bmod p$ is publicly computable and serves as a time metadata tag that does not provide confidentiality or integrity protection. Temporal access control is enforced through the access policy (a time attribute row in LSSS), protected by CP-ABE security. The CSP is assumed semi-trusted (honest-but-curious): it follows the protocol but may attempt to learn from ciphertexts. This assumption reflects IIoT deployments where the cloud provider

operates under contractual SLA and regulatory compliance, making fully malicious behavior (ciphertext tampering, token discard) unlikely but not impossible; defending against such attacks would require additional integrity mechanisms such as ciphertext signatures. Side-channel attacks, denial-of-service, and physical tampering are out of scope.

C. Security Model

We define IND-sCPA-T, extending standard CP-ABE security [4], [5] to capture interactions among temporal constraints, threshold policies, and revocation.

Exp_A^{IND-sCPA-T}: Our scheme guarantees IND-sCPA-T security if only a negligible advantage can be achieved during the interaction with any PPT adversary $\mathcal{A}$.

- 1) **Init**: $\mathcal{A}$ commits to target LSSS structure $\mathbb{M}^*$, temporal policy T_p^* , target time t^* , and revocation state R^* (a mapping from each attribute a to its current version v_a).
- 2) **Setup**: $\mathcal{C}$ runs $\text{SETUP}(\lambda)$ to generate (PP, MK) , gives PP to $\mathcal{A}$, and retains MK secret.
- 3) **Phase 1**: $\mathcal{A}$ adaptively queries:
 - a) **Key** (S_i, T_i, R_i) : returns SK for S_i under time policy T_i and revocation state R_i , subject to the restriction that (S_i, T_i, R_i) must not simultaneously satisfy $\mathbb{M}^*$, T_p^* at t^* , and R^* (all attribute versions current).
 - b) **Time token** t_j : returns the public token for period t_j .
 - c) **Revocation** (a_j, v_j) : increments a_j 's version.
- 4) **Challenge**: $\mathcal{A}$ submits M_0, M_1 . $\mathcal{C}$ encrypts M_γ ($\gamma \in_R \{0, 1\}$) under $(\mathbb{M}^*, T_p^*, R^*)$ at t^* and returns CT^* to $\mathcal{A}$.
- 5) **Phase 2**: Same queries as Phase 1 with same restrictions.
- 6) **Guess**: $\mathcal{A}$ outputs γ' .

The adversary's advantage is defined as

$$\text{Adv}_{\mathcal{A}}^{\text{IND-sCPA-T}}(\lambda) = |\Pr[\gamma' = \gamma] - \frac{1}{2}| \quad (3)$$

The model extends IND-sCPA with time token queries capturing forward security, revocation queries capturing version-based revocation, and challenge ciphertexts encrypted at a specific time under a specific revocation state. Selective security is standard for CP-ABE with enhanced features due to the exponential selective-to-adaptive reduction [4]; for IIoT deployments where policies are determined at deployment and remain static, this is practically sufficient.

V. OUR CONSTRUCTION

To address the security and functionality challenges of IIoT digital twin access control, we propose T-CP-ABE, a time-aware CP-ABE scheme that synergizes temporal predicates, THRESHOLD gates, and version-based revocation. The scheme operates as follows: The TA initializes system parameters and a master secret key. The encryptor obtains the current period's public time token from the TTA and embeds the period index in the ciphertext via a publicly computable binding factor. Authorized users decrypt ciphertexts whose attributes satisfy the LSSS access structure and whose time tokens fall within the authorized window. Revocation is enforced by incrementing attribute versions, which invalidates prior keys. The following describes the formal construction.

Algorithm 1 Encryption**Require:** $PP, M \in \mathbb{G}_T, \mathbb{A}, T_p$ **Ensure:** Ciphertext CT

- 1: Select $s \in_R \mathbb{Z}_p$; compute $C_0 = M \cdot e(g, g)^{\alpha s}$, $C_1 = g^{\beta s}$
- 2: Share s via Shamir over $\mathbb{A}$: for each leaf y , obtain share s_y
- 3: **for** each leaf y in $\mathbb{A}$ **do**
- 4: Compute $C_y = g^{s_y}$, $C'_y = H(\text{attr}_y \| v_{\text{attr}_y})^{s_y}$
- 5: **end for**
- 6: Obtain public time token (i, T_i) from TTA for current period i
- 7: Compute $b = \mathcal{H}_Z(i \| T_i) \bmod p$; update $C_0 \leftarrow C_0 \cdot e(g, g)^{\alpha \cdot b}$
- 8: **Output** $CT = (C_0, C_1, \{C_y, C'_y\}_y, \mathbb{A}, T_p, i)$

A. Core Algorithms

We employ a bilinear pairing $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ of prime order p with generator $g \in \mathbb{G}$, a hash-to-group function $H : \{0, 1\}^* \rightarrow \mathbb{G}$ modeled as a random oracle, and a hash-to-integer function $\mathcal{H}_Z$ (SHA-256 mapping to $\mathbb{Z}_p$) also modeled as a random oracle.

1) *Setup*(λ): Generates bilinear group parameters, selects $\alpha, \beta \in_R \mathbb{Z}_p$, and initializes the TTA hash-chain of length N . Setup samples $g \in \mathbb{G}$, $\alpha, \beta \in_R \mathbb{Z}_p$, computes $h = g^\beta$ and $e(g, g)^\alpha$, defines $H : \{0, 1\}^* \rightarrow \mathbb{G}$, and outputs $PP = (g, h, e(g, g)^\alpha, H, p)$, $MK = (\alpha, \beta)$.

2) *KeyGen*(PP, MK, S): Generates a private key for attribute set S with per-attribute randomness. No time token is stored in the secret key; time tokens are public values released by the TTA at each period. KeyGen selects $r \in_R \mathbb{Z}_p$, computes $K_0 = g^{(\alpha+r)/\beta}$, and for each $a \in S$ selects $r_a \in_R \mathbb{Z}_p$ and computes $K_a = g^r \cdot H(a \| v_a)^{r_a}$, $K'_a = g^{r_a}$.

3) *Encrypt*($PP, M, \mathbb{A}, T_p$): Encrypts message $M \in \mathbb{G}_T$ under access structure $\mathbb{A}$ and temporal policy T_p . The encryptor obtains the current period's public time token T_i from the TTA and embeds the period index i in the ciphertext.

4) *Decrypt*(PP, CT, SK): Recovers M if attributes and time satisfy the policy. The decryptor reads the period index i from CT , obtains the corresponding public token T_i from the TTA, verifies it against the published chain tip, and removes the binding.

B. Threshold Gates and Attribute Revocation

THRESHOLD(k, n) gates evaluate to true if and only if at least k of n children are satisfied. The secret s is distributed via a degree- $(k-1)$ polynomial over $\mathbb{Z}_p$ as described in Section III. Any k valid shares reconstruct s through Eq. (1). This mechanism subsumes AND gates ($k=n$) and OR gates ($k=1$) as special cases.

Attribute revocation operates at the version level. Each attribute a maintains a version v_a in public state R . KeyGen uses $H(a \| v_a)$ while Encrypt uses $H(a \| v_a)^{s_a}$, embedding version information in each ciphertext leaf node. Revoking attribute a increments v_a , causing keys generated under the old version to fail decryption since $H(a \| v_a) \neq H(a \| v'_a)$ for $v_a \neq v'_a$.

Algorithm 2 Decryption**Require:** PP, CT, SK **Ensure:** M or $\perp$

- 1: **if** current time t does not satisfy T_p **then**
- 2: **return** $\perp$
- 3: **end if**
- 4: Verify attribute versions: for each $a \in S$ in SK , check v_a matches current version in R ; if any mismatch, **return** $\perp$
- 5: Read period index i from CT ; obtain public token T_i from TTA
- 6: Verify T_i : compute $H^{N-i}(T_i)$ and check $= T_N$ (chain tip); if fails, **return** $\perp$
- 7: Remove binding: $C'_0 \leftarrow C_0 / e(g, g)^{\alpha \cdot b}$ where $b = \mathcal{H}_Z(i \| T_i) \bmod p$
- 8: Recursively evaluate $\mathbb{A}$: for leaf y with $a \in S$, compute $e(C_y, K_a) / e(C'_y, K'_a)$; for AND/OR gates combine children directly; for THRESHOLD(t_x, n_x) gates, collect $\geq t_x$ valid children and apply Lagrange (Eq. 1)
- 9: Let $A = e(g, g)^{r \cdot s}$ from root; compute $e(C_1, K_0) = e(g, g)^{(\alpha+r)s}$
- 10: **return** $M = C'_0 / (e(C_1, K_0) / A)$

C. Time Tokens and Binding

The TTA maintains a hash chain $T_0 = H(\text{seed})$, $T_i = H(T_{i-1})$, with tip T_N published as a public parameter. At each period i , the TTA publicly releases T_i , verifiable via $H^{N-i}(T_i) = T_N$.

The temporal access control itself is enforced through a time attribute row in the LSSS matrix $\mathbb{A}$: the encryptor embeds the current period index i as a policy row, and decryption succeeds only if the decryptor's time attribute satisfies this row. The hash chain provides forward security—given T_i , recovering any prior T_j ($j < i$) requires inverting $i-j$ successive hash evaluations, each costing $O(p)$ queries to the random oracle for a total of $(i-j) \cdot O(p)$.

During encryption, the encryptor also obtains T_i and computes $C_0 \leftarrow C_0 \cdot e(g, g)^{\alpha \cdot b}$ where $b = \mathcal{H}_Z(i \| T_i) \bmod p$. This binding records the encryption epoch as a metadata tag in C_0 ; it provides no confidentiality, integrity, or tamper resistance, and because $e(g, g)^\alpha$ is derivable from the public parameters, any party can re-randomize C_0 to substitute a different epoch. The period index i stored in CT already provides epoch identification for decryption; the metadata tag is a by-product of the hash-chain time token construction and does not independently contribute to security. To prevent time-tag tampering, the DO signs the full ciphertext (including C_0 and the period index i) with BLS during encryption, and the DU verifies this signature before invoking the ABE decryption engine. BLS verification costs a single pairing (≈ 2.3 ms), introducing negligible overhead while binding the ciphertext to its intended epoch. The BLS signature is thus an integral component of the time token mechanism rather than an external add-on.

We acknowledge that an adversary who can tamper with the period index i stored in CT could substitute a different epoch

without BLS protection; the BLS signature mitigates this by ensuring ciphertext integrity.

D. Security Proofs

The following theorem establishes the IND-sCPA-T security of T-CP-ABE.

Theorem 1. IND-sCPA-T Security: *T-CP-ABE is IND-sCPA-T secure under the DBDH assumption in the random oracle model. For any PPT adversary $\mathcal{A}$ making at most q_H random oracle queries, q_K key queries, and q_T time token queries, there exists a PPT simulator $\mathcal{B}$ such that*

$$\text{Adv}_{\mathcal{A}}^{\text{IND-sCPA-T}}(\lambda) \leq 2\text{Adv}_{\mathcal{B}}^{\text{DBDH}}(\lambda) + \frac{q_H + q_K + q_T}{p}. \quad (4)$$

Proof: We construct a PPT simulator $\mathcal{B}$ that uses $\mathcal{A}$ to solve DBDH. $\mathcal{B}$ receives $(g, A = g^a, B = g^b, C = g^c, T)$ and must decide whether $T = e(g, g)^{abc}$ or $T \in_R \mathbb{G}_T$. In Setup, $\mathcal{B}$ embeds $\alpha = ab$ by setting $e(g, g)^\alpha = e(A, g^b)$ and $\beta = b$ so that $h = g^b$. The random oracle H is programmed as $H(a\|v) = g^{u_{a,v}}$ where $u_{a,v}$ is chosen by $\mathcal{B}$ and recorded, so that $\mathcal{B}$ knows the discrete log of every hash output. For KeyGen, $\mathcal{B}$ uses the Waters partitioning argument to answer queries for attribute sets not satisfying $\mathbb{M}^*$. In Challenge, $\mathcal{B}$ sets $s = c$ and constructs LSSS shares; the binding $e(g, g)^{\alpha \cdot b_i}$ is computable as $e(A, g^b)^{b_i}$ since $\alpha = ab$. Time tokens are generated from independent random seeds. If $T = e(g, g)^{abc}$, the simulation is perfect; if T is random, C_0 is uniform. By a hybrid argument, the advantage bound follows. ■

Hybrid argument sketch. Define games G_0 (real) through G_3 (random): $G_0 \rightarrow G_1$: replace $e(g, g)^{abc}$ with random in C_0 (by DBDH); $G_1 \rightarrow G_2$: replace all hash responses with random group elements (by random oracle programmability, losing $\leq q_H/p$); $G_2 \rightarrow G_3$: simulate all key queries without MK using Waters partitioning (losing $\leq q_K/p$); time token queries are public and introduce no additional loss. The total advantage bound follows from a standard hybrid argument.

We now clarify the assumptions underlying the above proof and their implications for deployment. The reduction relies on three conditions: (i) the DBDH problem is hard in $\mathbb{G}$, which holds for the SS1024 curve at ≈ 128 -bit security; (ii) H and $\mathcal{H}_Z$ are modeled as random oracles; (iii) the adversary commits to $(\mathbb{M}^*, T_p^*, t^*)$ before Setup (selective security). Adaptive security—where the adversary chooses the challenge after seeing PP —remains open and would require a dual-system encryption argument. Furthermore, like all single-authority CP-ABE schemes, T-CP-ABE inherits key escrow: the authority holding $MK = (\alpha, \beta)$ can derive any user's secret key. Multi-authority extensions [20] can mitigate this by distributing trust, as discussed in Section VII.

The following lemmas establish complementary security properties.

Lemma 2. Forward Security: *Given time token T_i , recovering any prior token T_j ($j < i$) requires inverting $i-j$ successive hash evaluations, each costing $O(p)$ queries to the random oracle [48] for a total of $(i-j) \cdot O(p)$.*

Theorem 3. Pre-Release Unforgeability: *Before the TTA releases T_i , computing $b = \mathcal{H}_Z(i\|T_i) \bmod p$ succeeds with probability $\leq 2^{-256}$.*

Theorem 4. Revocation Soundness: *If attribute a is revoked ($v \rightarrow v+1$), keys with version v cannot decrypt ciphertexts under $v+1$ because $H(a\|v) \neq H(a\|v+1)$ in the random oracle model.*

Proof: With mismatched versions, the decryption pairing at leaf a computes $e(H(a\|v+1)^{s_a}, H(a\|v)^{r_a})$. In the random oracle model, $H(a\|v)$ and $H(a\|v+1)$ are independent random group elements, so this pairing output is uniform in $\mathbb{G}_T$. Lagrange interpolation at the root yields a random value, causing decryption to output $\perp$. The single-attribute proof extends to concurrent revocation: since each attribute's version hash $H(a\|v)$ is independent in the random oracle, revoking multiple attributes simultaneously yields independent uniform pairing outputs, each causing decryption failure at the corresponding leaf node. ■

Theorem 5. Threshold-Revocation Collusion Resistance: *In a THRESHOLD(k, n) gate, an adversary holding $k-1$ valid keys and one revoked key cannot reconstruct the root secret.*

Proof: Each leaf node ℓ in the THRESHOLD subtree is bound to $H(a_\ell\|v_\ell)$. A revoked attribute a_j at version $v'_j \neq v_j$ yields an independent random pairing output $u_j \in_R \mathbb{G}_T$ via the random oracle. The Lagrange interpolation at the threshold root computes $s = \sum_{j \in S} \lambda_j \Delta_{j,S}(0)$, where the share from the revoked leaf is uniformly random and independent of s . With probability $1 - 1/p$, the reconstructed value differs from the correct secret, causing decryption to fail regardless of the other $k-1$ valid shares. This holds for any subset of revoked attributes within the threshold gate, as each revoked leaf contributes an independent random element. ■

Experimentally, we verified collusion resistance across four scenarios (THRESHOLD with 50% revoked, AND with 50% revoked, all revoked, single revoked): all 20 trials produced decryption failure, confirming 100% resistance.

The security properties of T-CP-ABE are as follows. Data confidentiality reduces to DBDH via Eq. (4). Temporal security is independently guaranteed by the hash chain's one-wayness and the preimage resistance of $\mathcal{H}_Z$. Attribute revocation is sound due to version binding in the hash function inputs. Data integrity and non-repudiation rely on BLS signatures with EUF-CMA security under the CDH assumption [48], [49].

VI. PERFORMANCE ANALYSIS

This section evaluates T-CP-ABE across scalability, component ablation, comparison with state-of-the-art schemes, and real-world scenario validation.

A. Experimental Setup

We implemented T-CP-ABE in Python 3.9 using Charm-Crypto [50] on an Intel Xeon Gold 6248R (3.0 GHz, 128 GB RAM) with the SS1024 curve (≈ 128 -bit security). Each experiment was repeated N times ($N=100$ for ≤ 500 attributes,

TABLE II

SCALABILITY: KEYGEN/ENCRYPT SHOW MEAN (MS); DECRYPT SHOWS MEAN $\pm$ STD (MS). SS1024 CURVE, FULL SYSTEM (TIME TOKEN BINDING + THRESHOLD + DIFFUSION + DT).

Attrs	KeyGen	Encrypt	Decrypt
10 ($N=20$)	175	168	261 $\pm$ 6.1
100 ($N=20$)	1628	167	261 $\pm$ 7.3
1000 ($N=10$)	15887	165	259 $\pm$ 9.0
10000 ($N=5$)	178834	187	258 $\pm$ 3.9

$N=20$ for 1,000, $N=10$ for 5,000, $N=5$ for 10,000), reporting mean $\pm$ std with 95% CI. Decryption latency varies across tables due to different configurations: Table II uses the full system (10-attribute AND, time binding); the ablation analysis isolates components (4 attributes); real-world validation uses scenario-specific policies without binding.

B. Scalability

Table II presents the scalability results from 10 to 10,000 attributes. KeyGen grows linearly as $O(n)$, reaching 179 s at 10,000—acceptable for offline provisioning. Encryption remains within 164–187 ms regardless of attribute count, confirming negligible binding overhead. A key design property of CP-ABE is the decoupling between user key size and policy complexity: decryption cost depends solely on the number of policy attributes $|S_T|$ in the LSSS matrix, not on the total number of attributes $|S_U|$ in the user's key. Table II confirms this by fixing the policy at 10 attributes while scaling $|S_U|$ from 10 to 10,000, yielding stable decryption at 258–261 ms (CV $< 3.5\%$). For real-time IoT control commands, where policies typically contain fewer than 10 attributes, this ensures sub-300 ms decryption. Scaling the policy from 10 to 1000 AND-gates (fixed $|S_U| = 1010$) yields linear growth in both Encrypt (158.6 ms to 14.8 s) and Decrypt (248.0 ms to 23.6 s), confirming that constant decrypt time in Table II reflects the fixed 10-attribute policy. These costs are acceptable for offline batch processing of historical audit data.

We note that the hash-chain verification in Algorithm 2 (Step 6) computes $H^{N-i}(T_i)$ to confirm the chain tip T_N , contributing $O(N-i)$ hash evaluations to decryption. The experiments above use small chain lengths ($N = 5-20$), which is appropriate for the evaluation time window. In long-running IIoT deployments, operators should select period granularity (hourly, daily) so that N remains manageable; for example, hourly periods yield $N = 8,760$ per year, keeping early-period verification within a few milliseconds. Finer granularity (e.g., per-second) would increase N proportionally; extending Algorithm 2 with checkpoint-based verification to bound this cost is left as future work.

Ablation analysis isolates each component's marginal contribution. Removing the time predicate reduces decryption by 19.4 ms (hash-chain verification and binding removal); the diffusion model saves 10.9 ms; the digital twin module saves 9.6 ms. The minimal BSW07 core (91.8 ms) is the baseline. Marginal contributions (39.9 ms total) exceed the measured system overhead (22.0 ms, 24.0% over baseline) because components share group-initialization and caching costs.

TABLE III

THRESHOLD, REVOCATION, AND HASH CHAIN PERFORMANCE. SS1024, $N=20$ TRIALS, MEAN $\pm$ STD.

n	k	Encrypt (ms)	Decrypt (ms)
<i>Fixed $n=20$, varying k:</i>			
20	1	305 $\pm$ 10	34 $\pm$ 2
20	10	304 $\pm$ 5	245 $\pm$ 6
20	20	305 $\pm$ 8	479 $\pm$ 12
<i>$k=\lfloor n/2 \rfloor$, varying n:</i>			
5	3	82 $\pm$ 3	58 $\pm$ 3
10	5	157 $\pm$ 6	127 $\pm$ 3
20	10	305 $\pm$ 7	244 $\pm$ 7
<i>Batch revoc. ($n=20, k=10$): revoke < 0.1 ms; old-key fail 100%; new-enc 346–361 ms.</i>			
<i>Hash chain verify (no ckpt / with ckpt@1000): $N=10^3$: 1.3/< 0.01 ms; $N=5 \times 10^3$: 6.5/< 0.01 ms; $N=10^4$: 13.4/< 0.01 ms.</i>			

TABLE IV

THEORETICAL COMPLEXITY COMPARISON. $|S_T|$: POLICY ATTRS; $|S_U|$: USER ATTRS; N : HASH CHAIN LENGTH.

Scheme	KeyGen	Decrypt	CT Size	Key Size
BSW07 [18]	$(3 + S_U)E_G$	$(2 + 2 S_T)E_p + S_T E_{G_T}$	$(1 + 2 S_T)L_G + L_{G_T}$	$(2 + S_U)L_G$
Guo24 [26]	—	1S (outsrc)	$(5 S_T + 1)L_G$	$(2 + S_U)L_G$
Zhang24 [27]	$(2 + S_T)t_e$	$2t_e$ (outsrc)	$(2 S_T + 3)L_G + L_{G_T}$	$3L_G$
T-CP-ABE	$(2 + 2 S_U)E_G$	$(3 + 2 S_T)E_p + S_T E_{G_T} + (N-i)E_k$	$(2 + 2 S_T)L_G + L_{G_T}$	$(2 + 2 S_U)L_G$

E_G, E_{G_T} : exp.; E_p : pairing; S : scalar mult.; E_k : hash; t_e : mod. exp.; L_G, L_{G_T} : element sizes. Guo24 outsources decryption to a trusted proxy. Zhang24 uses IBE-to-ABE proxy re-encryption.

C. Threshold Policy Performance

Table III validates (k, n) THRESHOLD gate overhead. Encrypt time depends on n (each attribute incurs a pairing) and is independent of k . Decrypt time scales linearly with k because Lagrange reconstruction requires k valid shares. For typical IIoT policies ($n \leq 10$), both Encrypt and Decrypt remain under 260 ms.

D. Comparison with State-of-the-Art

We compared T-CP-ABE with BSW07, Guo24 [26], and Zhang24 [27] in terms of theoretical computational and storage complexity (Table IV).

T-CP-ABE's KeyGen overhead ($2.5-2.6\times$ vs. BSW07) stems from time binding and version computation—acceptable for offline provisioning. Guo24 achieves the lowest decryption cost (1S) via pairing-free outsourced decryption, but relies on a trusted proxy incompatible with our semi-trusted CSP model. Zhang24 incurs constant local costs through IBE-to-ABE proxy re-encryption but requires $2t_p$ cloud-side re-encryption per ciphertext. T-CP-ABE's decryption cost $(3 + 2|S_T|)E_p$ is comparable to BSW07's $(2 + 2|S_T|)E_p$ while providing temporal predicates, THRESHOLD gates, version-based revocation, and integrated threat detection.

Revoking attribute a_j requires incrementing v_j (< 0.1 ms) and re-encrypting affected components, averaging 164 ms per event at 1% monthly turnover. Batch revocation and hash chain verification results are consolidated in Table III.

Measured storage: SK grows linearly at 101 B per attribute (1.04 KB for 10, 9.92 KB for 100 attributes); CT core is constant 0.52 KB regardless of policy size. Compared with BSW07 on SS1024, T-CP-ABE SK is $\approx 2\times$ larger (extra time-binding and version-hash components) while CT is $3.9\times$ smaller at 10 attributes (0.52 vs. 2.02 KB) due to policy encoding differences, yielding a net bandwidth advantage for

policy-heavy deployments. Under 5G, transmission latency stays below 10.6 ms at 1,000 attributes.

E. Real-World Scenario Validation

We validate T-CP-ABE across four scenarios and UNSW-NB15. These experiments use static time attributes (validity period as LSSS row) without dynamic binding, isolating core CP-ABE plus time predicate overhead; full system overhead with binding is in Table II.

Decryption remains stable at 65–69 ms across all scenarios. The Industrial IoT match rate (20%) is consistent with the theoretical minimum $4/25 = 16\%$ for random AND satisfaction, confirming attribute-policy alignment rather than system failure. Cloud Storage achieves 40% due to its larger attribute pool.

For UNSW-NB15, we map 5,000 network traffic records to ABE attribute space (27 attributes, 7 attack categories). Across 3–15 attributes, all configurations achieve 100% functional correctness with KeyGen 103–185 ms, encryption 66–125 ms, and decryption 101–193 ms. P95 tail latency is 30–40% above the mean, confirming acceptable tail behavior for real-time IoT.

AI-driven closed-loop validation. We measure the end-to-end latency of the security response loop: anomaly detection trigger → attribute revocation → ciphertext re-encryption → new key issuance → old-key rejection. With 20 attributes and $k=10$ THRESHOLD policy, the full loop completes in 910 ms (revoke <0.1 ms, re-encrypt 325 ms, KeyGen 332 ms, Decrypt 252 ms). Old keys are rejected at 100% across all tested revocation counts (1, 3, 5 attributes), confirming that the diffusion-triggered revocation mechanism successfully blocks compromised users within a single policy update cycle.

VII. CONCLUSION

We proposed T-CP-ABE, a time-aware CP-ABE framework unifying temporal predicates, THRESHOLD gates, and version-based revocation with IND-sCPA-T security under DBDH. A lightweight metadata tag records the encryption epoch in C_0 ; access control is enforced through the LSSS policy. Experiments show that for latency-sensitive real-time control loops (typical policy ≤ 10 attributes), decryption meets the <300 ms deadline; for complex offline audits (up to 1000 policy attributes), the linear scaling to 23.6 s remains acceptable as a background batch job. KeyGen scales linearly across 10–10,000 user attributes. UNSW-NB15 and four real-world scenarios confirm practical applicability for industrial digital twins.

Nevertheless, several limitations exist. Experiments use software pairings on commodity hardware without IoT device evaluation. IND-sCPA-T requires selective security; adaptive security remains open and would require a dual-system encryption argument [19] that doubles group element size in ciphertexts and keys, increasing storage and computation by $\approx 2\times$ —a significant burden for resource-constrained IIoT. Selective security is adequate for industrial digital twin deployments where access policies are defined at system configuration time and remain static throughout the operational

lifecycle, as the adversary cannot influence policy selection. The time binding provides no confidentiality protection. The construction assumes a single authority, inheriting key escrow—the authority can derive any user’s key from MK . Attribute revocation provides forward security only; historical ciphertexts encrypted under earlier attribute versions remain accessible to revoked users with old keys, lacking backward security. Protecting historical data would require a ciphertext re-encryption mechanism, which is out of scope. In practice, industrial deployments can mitigate this gap through two compensation strategies: (i) *periodic batch re-encryption*, where the TA re-encrypts ciphertexts under the updated attribute versions during scheduled maintenance windows, and (ii) *TTL-based key expiry*, where each user key carries a validity period after which it is automatically invalidated, limiting the window of exposure for historical data. The diffusion model currently runs on cloud infrastructure; deploying it on edge devices requires model compression (knowledge distillation or quantization), which we leave as future work alongside adaptive threshold tuning to replace the fixed 95th-percentile rule. The BLS and diffusion components operate as system-level modules designed to support a feedback loop: BLS mutual authentication rejects unauthorized decryption requests before invoking the ABE engine, reducing wasted pairing computation; the diffusion model’s anomaly scores are designed to trigger the TA to update attribute versions v_a , closing the loop from passive encryption to active risk response. End-to-end automated evaluation of this feedback loop is future work. Future work follows three directions: (i) **Trust distribution**: multi-authority extensions [21], [23] with threshold secret sharing of MK to eliminate key escrow; (ii) **Post-quantum migration**: lattice-based ABE [28] with model compression for edge deployment of the diffusion component; (iii) **Backward security**: ciphertext re-encryption with batch update for historical twin data protection.

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